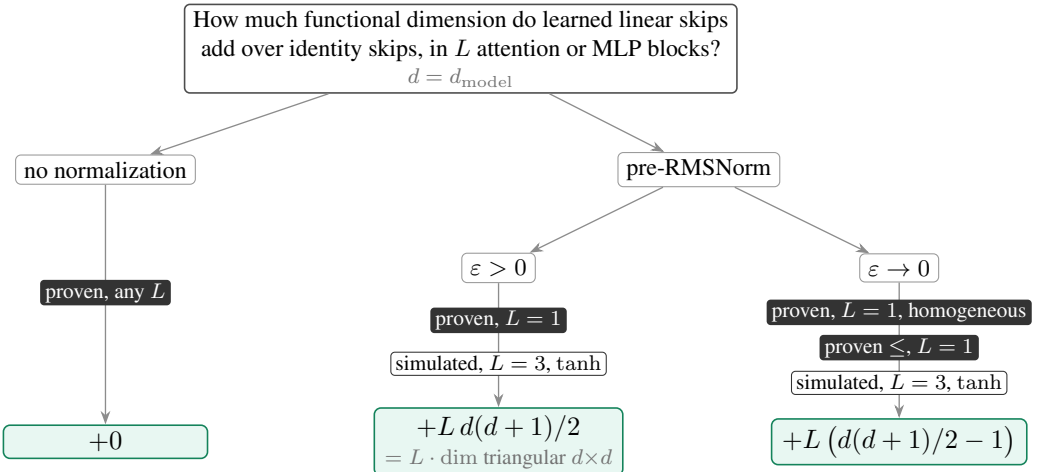

How much functional dimension do learned linear skips add over identity skips, in L attention or MLP blocks?
$d = d_{\text{model}}$
no normalization
pre-RMSNorm
proven, any L
$+0$
$\varepsilon > 0$
$\varepsilon \to 0$
proven, $L = 1$
simulated, $L = 3$, tanh
$+L\, d(d+1)/2$
$= L \cdot \dim$ triangular $d \times d$
proven, $L = 1$, homogeneous
proven $\leq$, $L = 1$
simulated, $L = 3$, tanh
$+L\left(d(d+1)/2 - 1\right)$